CFD ANALYSIS OF CIRCULATING FLUIDIZED BED COMBUSTION

Paper Summary

SUMMARY PASSAGE 1

I. Introduction

In the 2D CFB combustor have the 3 inlet points such as inlet for fluidizing velocity (primary air), inlet for coal particles and inlet for secondary air. The primary air is used for the fluidized of coal particle and secondary air for proper combustion in the combustor. The combustion processes is done by the discrete phase model and use the single injection system for burning of coal in the combustor.

SUMMARY PASSAGE 2

Ii. Literature Review

Having dynamic and parametric uncertainties in the model, a robust control algorithm based on linear matrix inequalities (LMI) have been applied to control the bed temperature by input parameters, i.e. coal feed rate and fluidization velocity. Circulating fluidized bed (CFB) combustion systems are increasingly used as superior coal burning systems in power generation due to their higher efficiency and lower emissions. L.X.

SUMMARY PASSAGE 3

V. Cfd Model Of Fluidized Bed

A circulating fluidized bed is one where fuel is burnt in a fast fluidized bed regime. In CFB boiler furnace the gas velocity is sufficiently high to blow all the solids out of the furnace. The majority of solids leaving the furnace is captured by gas-solid separators.

SUMMARY PASSAGE 4

Vi.Ii Grids Selected For The Present Work

If not then we have to change the solution parameters and sometimes solution method also. Currently, K-epsilon method is used for the combustion study of the fluidized bed. The boundary conditions are as equally important as the selection of the proper mathematical model.

SUMMARY PASSAGE 5

Boundary

Primary(Fluidized) Velocity inlet Inlet2

SUMMARY PASSAGE 6

Xiv.I Analysis Of Combustion At Fluidizing Velocity 4 M/S Total Temperature

In this figure, the bed and temperature are increasing as soon as the coal particles are burning and finally obtained the maximum value of total temperature after coal combustion is K. The value of static pressure is found in the combustor is (Pascal) after combustion process. The static pressure is rapidly increased when air fuel velocity enters in the combustion chamber. In the combustor, maximum static pressure is at the inlet points of the furnace.

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SUMMARY PASSAGE 8

Static Pressure

This is because with increase in air velocity and mass flow rate of fuel in the combustor increases thereby increasing the pressure across the combustor. In this fig.16,when the fluidizing velocity is 4m/s at the inlet of combustor then the static pressure is observed 5.63 Pascal after coal combustion in the combustor of CFB i.e. the static pressure is increases with increasing the inlet velocity. The same conditions are for total and dynamic pressure.

SUMMARY PASSAGE 9

Xviii. Conclusion

In this work, analysis of coal combustion in circulating fluidized bed has been performed at three different fluidizing velocities with fluent software. Following conclusions are drawn from the computational analysis in this present work.

SUMMARY PASSAGE 10

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In fluidized bed combustion, a number of parameters are important such as temperature, pressure, as it has been observed that all the parameters temperature and pressure are better for combustion at fluidizing velocity 6m/s. Therefore, a fluidizing velocity of 6m/s is suitable for fluidized bed combustion as compared to 4m/s and 5m/s.